

Structured Prompt■Driven Engineering (SPAD)

Fundational Skill Definition – Seachad

1. Purpose

SPAD defines a disciplined, auditable and repeatable methodology for designing, implementing and validating complex software and AI systems using structured prompts. The objective is to separate reasoning, planning, implementation and correction, minimizing technical debt and systemic risk.

2. Core Principles

- No implementation without an approved plan.
- No plan without an explicit audit.
- No corrections without a diagnosed root cause.
- Every output must be auditable.
- All changes must be justified technically.
- The methodology must be reusable across tools and LLMs.

3. Methodology Phases

Phase 0 – Context & Objective

Define the problem, scope, constraints and success criteria. This phase is driven by human intent and business context.

Phase 1 – PLAN

Architectural and logical design without writing code. Components, responsibilities, data flows and design decisions are defined explicitly.

Phase 2 – AUDIT_PLAN

Independent technical review of the plan. The design is evaluated for coupling, cohesion, scalability, concurrency, security and observability. A GO / GO WITH CHANGES / NO■GO decision is issued.

Phase 3 – CODE_PRIMER

Definition of implementation constraints, conventions, contracts and anti■patterns. This phase prepares the system for high■quality code generation.

Phase 4 – AUDIT_CODE

Verification that the generated code complies with the approved plan, respects architectural constraints and does not introduce hidden risks.

Phase 5 – FIX_PRIMERS

Targeted correction prompts focused on minimal, justified changes. Iteration continues until a stable and compliant state is reached.

4. Value for Seachad

- Creates a proprietary, auditable AI-assisted engineering process.
- Reduces refactoring cost and late-stage failures.
- Enables safe scaling of agent-based and LLM-driven systems.
- Provides a sellable methodology, not just tooling.
- Acts as a foundation for autonomous or semi-autonomous agents.

5. Usage Policy

SPAD is the default methodology for all Seachad projects involving AI-assisted design, development or decision automation. Deviations must be explicitly justified and approved.

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